

Brief Review ■

Natural Language Generation in Health Care

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Abstract Good communication is vital in health care, both among health care professionals, and between health care professionals and their patients. And well-written documents, describing and/or explaining the information in structured databases may be easier to comprehend, more edifying, and even more convincing than the structured data, even when presented in tabular or graphic form. Documents may be automatically generated from structured data, using techniques from the field of natural language generation. These techniques are concerned with how the content, organization and language used in a document can be dynamically selected, depending on the audience and context. They have been used to generate health education materials, explanations and critiques in decision support systems, and medical reports and progress notes.

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Effective communication is vital in health care, both between health care providers and their patients and among health care providers themselves. Different participants in the health care process—consultants, nurses, general practitioners, medical researchers, patients, their relatives, and even accountants and administrators—must all be able to obtain and communicate relevant information on patients and their treatment. But there are many obstacles in the way of effective communication: Participants may use different terms to describe the same thing—a particular problem for patients who do not understand medical terminology. Different participants frequently have different information needs and little time to filter information, so that no single report is truly adequate

for all. And the different participants may rarely have time to meet, yet the care of a patient is shared and passed between them.

The information required by different participants is increasingly available in coded and structured forms—in the patient record, in drug databases, and in knowledge bases of medical terminology. For population studies (in epidemiology and administration), having data in a coded and structured form enables precise queries to be formulated and quantitative analyses to be done on large bodies of data. Yet when data are being used to convince, justify, or explain, or to describe the status of a single individual, the more familiar medium of plain text and graphics may be more appropriate and effective: A focused coherent written report may be easier to deal with than the output from a set of database queries.

In health care, the evident need to translate between textual forms (human authored texts) and structured information has led to a large and continually growing body of research and development in natural language understanding.¹ In this article we consider the reverse problem—how textual documents may be produced from structured data. In particular, we show how a range of current natural language generation techniques can be used to produce from the same data, many different documents with different content, terminology and style, and thereby help meet diverse information needs within health care.

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Brief Overview of Natural Language Generation

Natural language generation (NLG) is concerned with automatically generating texts in English (or other human languages) from computer-accessible data. (Batesman and Hovy provide a recent review.²)* NLG techniques range from the simplest report generation and mail merge† systems, to sophisticated discourse and dialogue generation systems that reason about the effect various forms and presentational structures will have on their recipients. Simple mail merge techniques have been used in practice, as well as in various experiments on whether personalisation of material used in patient education³⁻⁷ can increase its effectiveness. However, the problem with such techniques is that they are inflexible, allowing relatively little variation in the texts produced. Attempts at allowing wider variation may result in texts that are no longer coherent, requiring post-editing by a person. Improving such systems requires more knowledge of language; more sophisticated NLG systems and techniques exploit research on human language processing. For example, NLG systems can exploit linguistic theories about where pronouns can be used and what they can be used to refer to, to automatically choose between using a pronoun or a full noun phrase at a particular point in a text. This might partially avoid the need for human post-editing in a report generation system.

Natural language generation may be divided into stages. One proposed division⁸ is as follows:

- **Text planning:** The basic content of the text is selected for the particular readership and organised coherently. Theories of text organisation may be used to find a good ordering of information.
- **Sentence planning:** The information is split into sentences and paragraphs, and appropriate use made of conjunctions, pronouns, etc.
- **Realization:** Grammatically correct sentences are produced. A grammar of the language (e.g., Eng-

lish) may be used, and knowledge of when different grammatical forms are appropriate.

Consider the task of producing a summary of the structured information in a patient record. Text planning methods would be used to extract and select the relevant information and decide on the basic ordering of that information. For example, the most recent information might be selected and ordered so that first all diagnoses and then all treatments are listed. Sentence planning methods would be used to divide that information into sentences, for example deciding to combine two separate facts (such as that the patient had a cough and that the patient has a sore throat) into one sentence. Finally, realization methods would be used to find a way to express the sentences in grammatical English: for example, as in the sentence, "The patient has a cough and a sore throat."

Not all systems will automate every stage. For example, current report generation and mail merge systems typically provide only for a simple form of text planning, and replace sentence planning and realization with simple "fill-in-the-blank" word strings. With respect to text planning, these systems allow information in a database or spreadsheet to control the selection of paragraph and sentence templates, but the basic form of each template must be provided by a human author. The system cannot automatically change the way facts are split into sentences or dynamically vary the grammatical forms or the terminology used.

More complex methods of text planning, sentence planning, and realization allow greater flexibility to be achieved. Terminology and style can be varied according to readership, and content can be selected to meet specific needs. The question that must be answered is how much flexibility is needed in a given application; if there is little variation in what the intended text is about or who is it for, then simple methods may suffice. However, the more variation in the intended audience, the more inherent complexity in the material to be presented; or the more the desire for communication to be a "two-way" interchange (i.e., a dialogue) rather than a one-time delivery of information, the more flexibility is needed. For many practical systems, only a limited amount of flexibility may be required—for example, where the texts to be produced have a standard structure (e.g., SOAP notes⁹), and the process of content selection involves filling out this structure. When only a small number of basic sentence forms are regularly used, it may be unnecessary to use a complex grammar to automatically generate sentences, as basic sentence templates (e.g., "Patient-X is suffering from Disease-Y") may be provided for all of the cases required.

*A more detailed and up to date survey is available online at web address: <http://www.cse.ogi.edu/CSLU/HLTSurvey/HLTSurvey.html>, chapter 4.

†Mail merge systems, available with most word processors, allow information from a database to be incorporated into a text document in simple ways, allowing, for example, mass production of personalised letters using information from a customer database. Simple IF-THEN statements often allow different chunks of text to be output depending on the current data.

For a particular application we have to decide whether the additional flexibility we get from using the more sophisticated methods justifies the costs; the more sophisticated methods do require more work in setting up for a particular application.¹⁰ This will depend on the kind of flexibility required for that application.

Example Applications

Within health care, nature language generation techniques have been applied in a number of areas: generating explanations, advice, and critiques in medical expert systems¹¹⁻¹⁸; generating reports, briefings, progress notes and discharge letters for health professionals¹⁹⁻²³; and generating explanatory materials for patients.²⁴⁻²⁸ This review considers each of these areas, considering the applicability of different language generation techniques in each.

Generating Expert System Explanations and Critiques

If the recommendations of an expert system or decision support system are to be understood and assessed by health professionals, then some explanation of the reasoning or rationale behind the recommendation should be available. Generating such explanations is an NLG task—a text is generated from computer-accessible data concerning the systems reasoning.

In the earliest work on explanation in rule-based expert systems, very simple NLG methods were viewed as sufficient, as the input was very constrained (just the trace of the system's reasoning), and relatively little variability in the output was required. Text could be generated to explain how the system reached a conclusion or why the system was asking a particular question. The text consisted of an English sentence whose clauses corresponded to the antecedent and consequent clauses of the rule from which a conclusion was drawn or that motivated a particular question being asked. No text planning was necessary, as only a single sentence was produced, and no sentence planning was necessary, since what was included in the sentence to be generated was fixed by the set of clauses in the rule. Realization consisted of stringing together text using simple templates associated with expert system rules or primitive functions. Early systems such as MYCIN showed how far this basic approach could be taken.²⁹

The explanations produced using these simple methods are far from ideal. However, improving them requires, as a basis, a richer source of information—a

simple rule trace will often miss key information that should be included in order for an explanation to be convincing. Swartout¹¹ attempted to address this issue by looking at how the underlying rationale behind rules may be preserved and used in explanation. Although only simple generation methods were used, this work is important in reminding us that one cannot get a good text without good data and that to improve the output of a text generation system, the first thing needed may be to improve the input.

Explanations were based on richer input in proto-type systems developed by Langlotz³⁰⁻³² and by Jimison.^{33,34} Langlotz's QxQ system was developed to demonstrate that the clinical use of quantitative decision models could be facilitated through generating non-quantitative explanations of their results—explaining which decision option was preferable, the basis for that preference, and the sensitivity of that preference to uncertainty in the relevant probabilities—using both text and graphics.

In producing its explanations, QxQ used a basic form of text planning; it had a small set of strategies modeled on the form and content of published medical decision analyses that were viewed as presenting persuasive arguments. QxQ used heuristic rules to select strategies that effectively used the available data to argue for the results of the model, and then merged the data with the selected frameworks to produce symbolic expressions that justify the difference in expected utility. QxQ's explanations required no sentence planning, as the type of material to be included in each sentence was specified in the strategies, and it used the same realization method as in the earlier MYCIN work. Since QxQ's explanations were longer and more complex than MYCIN's, more of the problems of inflexible realization were apparent in QxQ's explanations. For example, in explaining why intermittent pneumatic compression prophylaxis is strongly indicated in a case of deep vein thrombosis, QxQ produced the following sentence:

"The decision is supported by the fact that the probability of deep vein thrombosis with no prophylaxis is greater than the probability of deep vein thrombosis with intermittent pneumatic compression prophylaxis."

If QxQ's realization strategies could make use of demonstrative pronouns and contextual abbreviations, this sentence could be more simply realised as:

"This decision is supported by the fact that the probability of deep vein thrombosis with no prophylaxis is greater than that with IPC prophylaxis."

Although intubation of this patient was not proposed, it is clearly desirable. Not intubating this patient would have the risk of aspiration.

Looking at the other aspects of the proposed plan, for a patient with chronic renal failure, Curare is a reasonable selection since it is reliably metabolized by the liver, and Halothane is a good choice since it has no nephrotoxicity.

Figure 1 Example Text From ATTENDING critiquing system.

Jimison's system produced explanations that incorporated patient-specific characteristics (such as their occupation, leisure activities and past experience of pain), as well as clinical factors.^{33,34} These personal characteristics influence patient preferences for different outcomes, and hence the choice of treatment. Jimison observed that texts which present decision models implicitly refer to additional variables such as these, yet these don't appear in the decision model. She showed how adding such variables explicitly to the model, along with distributions on their range of possible values, could be used to produce explanations that could contrast the recommendation for a specific patient with that for the typical or generic patient. This allowed the physician who was given the explanation to understand the importance of each variable to the particular decision. The methods used for producing texts were similar to that used in QxQ. While both QxQ and Jimison's system remained prototypes, both stand as significant proofs of concept.

Because QxQ had only a small set of presentation strategies, it had no need for more complex text planning—for example, to mediate between strategies when more than one applied. More flexible and richer explanations may be generated by treating text or explanation planning as an independent problem. This is discussed by Moore³⁵ (although not for a medical domain). A simple example of independent text planning in a medical expert system is HF-EXPLAIN,¹² which generates explanations for a heart failure expert system based on a causal model. Rather than merely providing a trace of the causal processes, the system tries to follow the typical structure of human explanations in this domain by making use of simple schemata.

Turning from the generation of explanations to that of critiques, a critiquing expert system comments on the user's suggestions rather than generating its own. Such a system must therefore generate a coherent critique based on its analysis of the user's proposal. For

example, ATTENDING analyses the risks and benefits associated with a proposed anaesthesia plan, and generates an English critique.^{13,14} The basic content and structure of this critique is fairly rigid—there is no explicit text planning stage. However, the way a critique is expressed requires flexibility because of the complexity of ATTENDING's analysis of the user's plan: simple template-based approaches would not be sufficiently flexible to produce fluent output. ATTENDING uses a slightly more complex realisation method for its generation system (PROSENET) based on an augmented transition network labelled with fragments of English. Traversing a particular network should result in a grammatically correct utterance whose details depend on contextual factors and details of the input. An example fragment of output is given in Figure 1.

Recent work on medical decision support has started to consider the issue of how evidence-based guidelines may be made widely available. In a computer-based guideline system, although the underlying reasoning may be simple, there is still the question of how best to present the guidelines (and patient specific advice) is the health professional. This has been explored by Barnes and Barnett³⁶ and by Day et al.³⁷ Barnes and Barnett, for example, use an approach similar to PROSENET to produce coherent patient-specific guidelines.

Where many guidelines are simultaneously active however, such simple presentational methods may

- Caution: get a chest x-ray immediately to rule out a simple right pneumothorax.
- Caution: get a chest x-ray immediately to rule out a simple right hemothorax.
- Do not perform local visual exploration of all abdominal wounds until after getting a chest x-ray. The outcome of the latter may affect the need to do the former.
- Please get a chest x-ray before performing local visual exploration of all abdominal wounds because it has a higher priority.

Integrated Critique Produced by TraumaGEN:

- Caution: get a chest x-ray to rule out a simple right pneumothorax and rule out a simple right hemothorax, and use the results of the chest x-ray to decide whether or not to perform local visual exploration of all abdominal wounds.

Figure 2 Critiques Produced by TraumaGEN.

not be sufficient. In such cases, a text planner must be able to take an arbitrary set of communicative goals, each relating to a different piece of advice, and produce text that expresses the entire set both concisely and coherently. This is done in TraumaGEN,¹⁸ which has been designed to produce coherent critiques in initial definitive management of multiple trauma. TraumaGEN works on the output of Trauma TIQ,^{16,17} which produces individual critiques of physician orders (or the lack thereof) based on what it infers to be the physician's current plan and on the recommendations of its associated expert system, TraumAID.³⁸ TraumaGEN addresses the problem that, while in isolation each of TraumaTIQ's noted critiques may effectively warn a physician about a problem, usually several problems are detected simultaneously, producing multiple critiques whose aggregation can be confusing (Fig. 2). TraumaGEN takes the set of individual critiques and integrates them into a more concise, coherent, and thereby more effective, form.

Generating Patient Information Materials

There are many reasons why patients should be given better information: to reduce patient anxiety, to enable patients to share in management decisions, to enable and encourage them to change their behavior (e.g., to stop smoking), to enable and encourage them to comply with treatment, to enable them to manage chronic conditions, and simply to increase patient satisfaction. Improved patient education through better information has the potential to save large amounts of time and money as well as lead to more satisfied and healthier patients. There is a large literature on the subject, but a good overview is given by Ley.³⁹

Currently, patient education is largely provided through verbal interaction with health professionals and through leaflets, posters, and other printed material. However, health professionals have limited time (and are not always good communicators), and generic leaflets are impersonal and unspecific, not addressing a particular patient's particular needs. Patient education researchers recognize the need for more personalized materials,⁴⁰ and simple mail merge techniques have been used effectively to produce personalized leaflets,³⁻⁵ which appear more effective than general ones. Simple interactive systems have also been shown to be acceptable to patients.^{41,42} Patients may be less embarrassed asking questions of a computer than of their doctors.⁴³

Although it is possible to produce personalized materials (leaflets and interactive systems) using fairly simple techniques, it is an area where NLG techniques can lead to more flexible systems and more coherent,

```
(define-text-plan-operator
:name alleviate-fears-female-pre-menopausal
:effect (alleviate-fears ?patient (forever ?disease))
:constraints
  ((female ?patient)
   (pre-menopausal ?patient)
   (not (in-patient-history? estrogens)))
:nucleus ((BEL ?patient
  (improve ?disease (after menopause)))
  (BEL ?patient
  (improve ?disease aging))))
```

Figure 3 Example Text Planning Operator in Migraine.

fluent texts. Several recent projects (Migraine, Piglit, OPADe, and HealthDoc) have explored this. In each, a text-generation system has been used to produce leaflets or interactive materials, based on a combination of (a) data from drug databases and/or medical knowledge bases, and (b) data on the specific patient. Information is selected to be included in the text, based on the patient's needs, including information specific to that patient.

The Migraine project^{24,25} is concerned with generating interactive materials for migraine patients. Screens of text are generated using an NLG system, with the user able to ask follow-up questions via a mixture of hypertext and menu selection. A fairly sophisticated text planner³⁵ is used to select content appropriate to each individual. The approach allows responses to later questions to be answered in the context of previous replies, referring to these earlier responses and avoiding excessive repetition.⁴⁴ The medical knowledge base required was constructed using the UMLS semantic network,^{45,46} and patient data were obtained through an online interview.

Text planning in Migraine proceeds by using text planning operators to expand goals into subgoals, depending on constraints, until a subgoal can be conveyed using a simple phrase or sentence. A sample text planning operator for Migraine is given in Figure 3. It states that one way to alleviate a female premenopausal patient's fears about the disease in question is to get her to believe that the disease improves after menopause and with aging. Given such a goal of alleviating patient fears, this plan operator could be used to expand that goal for an appropriate patient, eventually producing a phrase or sentence aimed at getting her to believe that the disease improves after menopause and one aimed at getting her to believe that the disease improves with aging. This goal-directed method of selecting text content has proved

BEZAFIBRATE

Bezafibrate is a **cardiovascular drug** which reduces the amount of some kinds of fat in the bloodstream. According to **your record** you are currently undergoing this treatment. It is often used to treat **hyperlipidaemia**. It could have some side effects, in particular **nausea**. Your prescription of bezafibrate comes in 200 mg tablets. It is to be taken three times each day.

YOUR RECORD MORE INFO HELP
Discuss with doctor?

Figure 4 Example Text from the PIGLIT system.

more flexible than methods used in mail merge and similar software.

Piglit²⁷ was similar to Migraine in many ways, using text planning methods to generate hypertext explanations for diabetes patients. Piglit, however, had closer links with the patient record, enabling the patients to explore topics mentioned in their records, while the record was used in turn to determine how these topics should be explained and to add personal reminders. Text planning was kept simple, with planning operators specifying the information that should be included when describing a particular class of medical concept (e.g., when describing a disease, describe its symptoms and possible treatments). Information relevant to a specific patient could be added (e.g., if the patient has this disease, describe its onset and how it is being treated). The medical knowledge base was constructed based on the organization of concepts in the Read coding scheme (used by the National Health Service in the UK) but has since been modified to use the GRAIL representation language from the GALEN medical terminology project.⁴⁷ Figure 4 gives an example text from the original system (words in bold can be clicked on for further information).

Both Piglit and Migraine use simple surface realization based on selecting from human-authored templates (e.g., "The symptoms of Disease-X are Symptoms-Y") and filling in with specific data. They have both resulted in working systems that have been evaluated with a small number of patients. A much larger randomized trial is now in progress, looking at the benefits of personalized information for cancer patients, using a system based on Piglit.⁴⁸ A recognized problem with both systems is the effort required to author the knowledge base (containing the general medical information that may be communicated to the

patient). However, once a suitable knowledge base is constructed, it appears easier to maintain it (as medical knowledge changes) than to maintain materials in a textual form. A particular fact (say, the drug recommended for a particular disease) will only appear once in the knowledge base, but it may be mentioned many times in textual documents, using many different surface forms.

The OPADE project looked at generating personalized leaflets about drugs.²⁶ As in Piglit and Migraine, the emphasis was on text planning. A notable feature of OPADE was that the knowledge to be communicated all came from existing resources (a drug database and the prescription), so authoring or adapting a knowledge base was not required. OPADE also attempted to address the problem of the potential conflict between what the doctor wants to communicate and what the patient wants to know.⁴⁹

The most recent project in this area is the HealthDoc project.²⁸ HealthDoc takes an unusual approach to text generation. Rather than generating text from a knowledge base, HealthDoc starts with a master document. Generation involves selecting from this master document and repairing the result to produce a coherent document. For example, when material is cut from the master text, pronouns (e.g., "it") may be left without a referent. Repairing a text might involve replacing one or more of these pronouns with a full noun phrase (e.g., "the angina pectoris").

The HealthDoc approach thus places most of its effort at the sentence-planning stage and may prove an effective practical technique when there is a relatively small amount of material to be selected from and when existing report generation or mail merge software results in texts that require significant post-editing by a human.

The above systems also differ in the techniques that were used to elicit system requirements—the kind of material patients require, and how material can best be adapted for an individual patient. In the Migraine project, ethnographic studies were used with patients.⁵⁰ In OPADE, surveys and questionnaires were used to find which topics patients considered important and what ordering of information was preferred,⁵¹ as well as interviews with health professionals, who were asked to explain topics to hypothetical patients.²⁶ Piglit²⁷ used (primarily) questionnaire-based feedback on early prototypes but also interviews with health professionals.

Generating Reports and Progress Notes

Another area in which NLG techniques have been tried experimentally in health care is in the generation

of discharge summaries and progress notes. When the care of a patient is shared among or passed on to other health professionals, it is important that a clear and accurate account is presented of the current state of the illness and treatment. Yet, producing such accounts is time consuming,⁵² and the required clarity demands both fluency and coherence.⁵³ Computer-based patient record systems may already have simple report generation facilities built in, which can be used to assist the health care professional. Because the resulting output may lack fluency, requiring significant post-ending, several researchers in medical informatics have looked at ways of generating better reports.^{19–21,54}

Generating better reports does not appear to require sophisticated text planning methods, as the structure of a progress note or discharge summary is generally fairly constrained. Indeed, the requirement for a consistent, easy-to-scan format may mean that too much variability in the output is undesirable. However, merely stringing together data in a sequence of short sentences or clauses will result in a report that lacks fluency and is overly verbose. Here, what is required are sentence planning techniques, especially ones that deal with aggregation (i.e., combining and merging clauses into concise sentences) and with pronominalization, both of which reduce repetition—hence, reducing verbosity—and increase coherence by bringing together related material. Simple methods can normally be used for realization, as frequently only a small number of basic sentence forms are needed.

The IVORY system,²¹ for example, was designed to generate textual progress notes given data entered by the physician. User-centered design methods were used to try to ensure that the system met the needs

SUBJECTIVE:

Constitutional:

Mild frontal headaches for the last 2 days. No fever and no chills.

ENT:

Constant, moderate sore throat for the last 1 day. The sore throat is worsening. No nasal discharge.

OBJECTIVE:

Vital Signs:

Oral temp: 99.8 F. Right brachial pulse: 89 sitting. Right upper arm bp 130/85 sitting.

Figure 5 Fraction of Progress Note Generated by IVORY.

The angina pectoris is progressive and moderate. It has an aching, burning, possibly intermittent character, is aggravated by cold and movement and relieved by rest. There is a cough and no breathlessness.

The cough is improving and mild. It has a dry, non-productive character and a bovine, harsh sound character. It is aggravated by dust and smoking and relieved by rest.

Figure 6 Example Report from Pen & Pad Reporter.

of physicians, and the format of the notes followed that used by physicians when writing progress notes by hand. In this case the SOAP format was used: first, subjective data are given (patient's reported symptoms); then, objective data (physician's observations and measurements); next, the physician's assessment; and finally, the physician's plan for treatment or further tests. While text planning in IVORY simply involved filling out the basic SOAP template with the relevant data, significant attention was paid to sentence planning, particularly combining clauses as a way of avoiding repetition. As a simple example, rather than generating "Three day history of cough. Three day history of sore throat," IVORY would generate "Three day history of cough and sore throat."

An example fragment from a progress note generated by IVORY is given in Figure 5. IVORY does not generate fully grammatical English sentences, but rather abbreviated phrases, similar to those used by many physicians and nurses. (The style is, in fact, rather standard "telegraphic" American English, which can be observed in a wide range of applications and is not specific to health care.) The structured progress note form is preferred by many health professionals, allowing easy access to relevant parts of the report. As the telegraphic style used in IVORY employs rather short constructions, a sophisticated grammar-based realization component was not seen to be required.

A similar system has been developed recently¹⁹ for generating summaries from the Pen&Pad patient record system.⁵⁵ While most work in this area focuses on the specific needs of the application, this work specifically considers NLG issues. Text planning is based on simple schemas⁵⁶ that provide a flexible way to structure the text. Sentence planning includes aggregation and pronominalization methods, and a realization module is based on a simple grammar, producing grammatically well-formed output (Fig. 6).

Other report-generation applications have used more sophisticated realization modules, corresponding to

more variability in the input. For example, Abella et al.²² describe a system for generating reports on radiographs, using a fairly sophisticated off-the-shelf sentence planning and realization system, FUF,⁵⁷ to produce sentences. Bernauer et al.²⁰ argue that for producing reports on bone scan studies, the variety of possible kinds of observations requires a relatively sophisticated realization module; they use one developed specially for their task. It appears that for generating medical reports there is no one generation architecture perfect for every application, but each specific application may make different demands and require different sorts of flexibility and variability to be supported.

The work mentioned above has been concerned with generating a written textual report. However, some information can be conveyed more effectively through graphics, while the use of spoken (rather than written) text can allow someone to attend to both text and graphics (or to both text and the world outside) simultaneously. This has led to a recent interest in multimedia generation. Generating a multimedia presentation from data involves selecting the appropriate modality (text, graphics, etc.) for communication and coordinating the presentation so that, for example, images are referred to in the text. Recent work at Columbia²³ has considered how multimedia postoperative briefings can be generated that meet the different needs of nurse and physician.

Generating Descriptions of Medical Concepts

In most of the above applications there is a need for generating descriptions of medical concepts given a standard code (e.g., SNOMED, ICD-9) for that concept. The complexity of this problem depends on the coding scheme used. In a system such as ICD-9, where a separate code is given for every distinguishable concept, then a simple mapping of code to a preferred English phrase may be possible. But in more complex compositional schemes, such as that used in the GALEN terminology server,⁴⁷ generating the appropriate English phrase for a compositional concept becomes more difficult.^{58,59} However, although the generation of noun phrases is more complex, the approach can pay off if multilingual output is required.⁵⁷ While the 1997 version of the UMLS⁴⁶ correlates terminologies in Spanish, German, Portuguese, and French with those in English, the mapping of terms is only partial; there are only translations available for all these languages for MeSH terms. For these languages and others, entering a preferred phrase by hand for each uncovered concept will require extensive effort.

In a compositional scheme, on the other hand, much can be achieved based on combining words from a relatively simple lexicon.

Discussion

The above systems have used a wide range of natural language generation (NLG) techniques, from the very simple to the complex. Some have emphasized text planning^{11–12,24–27,18,1}; others have emphasized sentence planning^{19,21,28} or realization.^{13,14,59} In most cases, those using more sophisticated techniques^{11–15,18,19,23–28,59} are research prototypes; these, while aiming to address a real need, have not been used in practice beyond small-scale evaluations. Here we briefly consider both where more advanced NLG techniques are likely to be worthwhile in practice and which methods are likely to be most useful.

Sometimes the costs of using advanced NLG techniques may outweigh the benefits.⁹ For example, if using them requires authoring a special-purpose knowledge base whose life span is short, then (unless the techniques allow one to serve a significantly larger population than one otherwise could) simple techniques such as those used in mail merge systems may be adequate. One useful compromise, which can be seen even in the research systems discussed earlier, may be to use advanced techniques for part of the process and simpler techniques for the rest. For example, simple text planning can be combined with sophisticated sentence planning and realization methods if good fluency is required or depending on context. Alternatively, simple fill-in-the-blanks sentence templates can be combined with sophisticated text planning methods if the selecting and structuring of the content are quite complex.

Clearly, whatever techniques are used, for a system to be used in practice it is important that it is integrated both with existing clinical systems (patient records, drug databases, etc.) and also with existing practice. This point has been frequently made for medical expert systems and applies equally to NLG systems used in health care. A system that gave “added value” to an existing patient record system would be more persuasive than a stand-alone system requiring separate or idiosyncratic data entry. Research prototypes may not go as far as using actual patient record systems, but they should at least take account of the medical coding schemes in use in such systems.

Conclusions

In health care communication, there is a real need for the generation of textual reports and explanatory ma-

terials from structured data. Natural language generation is a rapidly maturing field. As techniques become better understood and more off-the-shelf tools become readily available, NLG offers real potential for better health care communication, increasing the flexibility and adaptability of systems and the fluency of output texts. There will always be a cost associated with using more sophisticated techniques, and these costs must be weighed against the benefits. However, intermediate techniques, using simpler techniques for part of the process, may provide ways to get maximum benefit for a particular application with minimal cost. The appropriate techniques will depend on the kind of flexibility and the style of text required for the particular problem.

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